

SANDEEP N MENON

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EXPERIENCE

Software Engineer, Perception | [Kodiak Robotics](#)

June 2022 – Present

- Owns end-to-end development of the image segmentation model, overseeing data labeling, quality assurance, model training, deployment and real-time monitoring and failure detection.
- Developed a synthetic weather data generation engine using diffusion models, enhancing model generalization across diverse environmental conditions, including snow and dirt tracks.
- Designed and implemented a multi-modal deep learning network taking in Camera, Lidar, and Radar sensor data for Birds Eye View(BEV) Segmentation and ground height estimation.
- Increased obstacle segmentation model accuracy by 20% using self-supervised masked autoencoders.
- Deployed inference engine to run 3D sparse convolutions and custom CUDA kernels under 30ms, 85% speedup from PyTorch. Facilitates fast model inference on onboard infrastructure of autonomous trucks.
- Collaborated with Kognic, Kolena, ClearML, Embedl and Ambaralla to add ML operational capabilities for data labeling, model training infrastructure and optimization.

Deep Learning Research Engineer | [Deepen AI](#)

Sept 2020 – Jul 2022

- Developed a 3D PointNet model for temporal smoothing of segmentation predictions over point cloud sequences, achieving a 20% mIoU improvement.
- Built a Sparse Point-Voxel CNN model for semantic segmentation of 3D point cloud sequences, boosting data annotation speed by 30% and obtaining a 76% mIoU score.
- Implemented 2D object-aware anchor-free tracking for auto-labeling, increasing labeling speed by 50%.
- Devised a targetless Camera-IMU and [stereo camera calibration](#) algorithm, reducing calibration time by 90% and achieving sub 1-degree error margin.
- Created on-demand GPU Virtual Machine allocation system saving up to 4000 USD/month for the company.

Software Development Engineer II | [Microsoft](#)

Jun 2018 – Sept 2020

- Co-authored new Machine Learning algorithm inspired by Random Forests to identify similar customer deals and opportunities in Dynamics 365's [Relationship Analytics](#); **received patent award**.
- Shipped [Dynamics 365 sales insights connector](#) in Power platforms managing 9 million monthly requests.

PROJECTS

[Federated Training System for Generative Adversarial Networks \(GANs\)](#) | *PyTorch, Flower*

Oct - Dec 2022

- Designed a federated learning system to train GANs across dozens of devices without sharing local data.

[Asymmetric 3D Convolutions in Torchsparse](#) | *PyTorch*

Feb 2021

- Contributed Asymmetric 3D Convolutions implementation to [TorchSparse](#) library, managed by MIT HAN Lab.

TECHNICAL SKILLS

Deep Learning: PyTorch, Tensorflow, TensorRT, Onnx, Keras, MMDet, PointNets, CNN, VAE, GAN

Optimization: CVXPY, Ceres, SciPy, NumPy

Computer Vision: OpenCV, LiDAR, SLAM, Multi-Sensor Calibration and Fusion, Depth Estimation

Languages/Platforms: C++, C#, Python, JavaScript, RUST, CUDA, OCaml, React, Redux, Django, LangChain, Flower, Bazel, ClearML, Embedl, Docker, Modal, Kolena, Azure, ROS, ClearML, WandB, Google Cloud, MongoDB, RocksDB, MySQL

EDUCATION

New York University (NYU) Courant Institute of Mathematical Sciences

2022 – 2024

Master of Science in Computer Science

National Institute of Technology Karnataka, Surathkal, India (NITK)

2014 – 2018

Bachelor of Technology in Computer Science